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Perspective

Two roads diverged: Pathways toward harnessing intelligence in neural cell cultures

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THE BIGGER PICTURE *In vitro* neural cell cultures are increasingly being used as a substrate to explore information processing and rudimentary forms of learning. More advanced systems have even been conceptualized as future intelligent devices with unique properties that may exceed the capability of silicon computing alone. This perspective introduces a conceptual framework for understanding two distinct but complementary approaches to achieving this goal: organoid intelligence (OI) vs. bioengineered intelligence (BI). While both broadly fall within the wider definition of synthetic biological intelligence (SBI), OI and BI diverge in their methods, design goals, and implications. Here, the advantages and disadvantages of each approach are analyzed with respect to the existing challenges and future opportunities. OI is a top-down approach that aims to recapitulate physiologically relevant brain structures through the use of organoid cultures. OI is closely tied to biological realism and seeks to emulate *in vivo* brain complexity, with the potential to advance neuroscience, biomedicine, and therapeutic development. Yet, limitations in interacting with organoids and the inherent complexity of these systems raise challenges in establishing reproducible intelligent devices. BI, in contrast, is proposed here as a primarily bottom-up approach that abandons physiological fidelity to assemble modular and highly controllable neural cultures that may maintain only a passing resemblance to the geometry of *in vivo* brains. BI involves constructing circuits of neural cells in highly specified ways through the principles of bioengineering, to assemble controllable and perhaps even task-optimized information processing and intelligent devices. While these two paradigms should be seen as points on a spectrum, explicitly defining and distinguishing OI and BI enables better communication, research coordination, and responsible innovation in the emerging field of intelligent *in vitro* neural systems. Finally, future applications at staged time points, along with the corresponding ethical considerations for these technological innovations are also discussed.

SUMMARY

Interest in using *in vitro* neural cell cultures embodied within structured information landscapes has rapidly grown. Whether for biomedical, basic science, or information processing and intelligence applications, these systems hold significant potential. Currently, coordinated efforts have established the field of organoid intelligence (OI) as one pathway. However, specifically engineering neural circuits could be leveraged to give rise to another pathway, which is proposed to be termed bioengineered intelligence (BI). This perspective examines the opportunities and prevailing challenges of OI and BI, proposing a framework for conceptualizing these different approaches using *in vitro* neural cell cultures for information processing and intelligence. In doing so, BI is formalized as a distinct innovative pathway that can progress in parallel with OI. Ultimately, it is proposed that while significant steps forward could be achieved with either pathway, the juxtaposition of results from each method will maximize progress in the most exciting, yet ethically sustainable, direction.

INTRODUCTION

When it comes to creating an intelligent device, the optimal materials and their ideal forms remain unknown. Recent research on silicon-based computing has rapidly accelerated to the point where it is considered the default. Driven by enormous investment in large language models (LLMs) and related approaches, interest in artificial intelligence (AI) has been galvanized to significantly impact worldwide economics.[1] With the promising applications of these AI technologies, there is rapidly growing interest in what improvements might theoretically yield an artificial general intelligence (AGI).[2,3] Terms such as AI and AGI lack a single, agreed-upon definition—an issue requiring a coordinated effort to address[4,5]—but broad distinctions are sufficient for the purpose of this paper. AI has previously been abstractly proposed as the maximal capacity of an artificial system to successfully achieve a novel goal through computational algorithms.[6] In contrast, AGI has been proposed to extend beyond AI systems' ability to perceive, possess knowledge, understand, learn, and function either partially or completely autonomously in a variety of environments and tasks.[6] In short, any colloquial understanding of AGI mirrors the capability displayed by biological intelligence, which use neural systems to process information. Indeed, a degree of generalized intelligence can be displayed by a range of organisms from insects to humans.[7] As such, while currently commercially available AI tools rely nearly exclusively on classical von Neumann computing architecture,[8] alternative substrates to developing a generalized intelligence system (artificial or otherwise) exist. Indeed, currently, biological systems, driven by neural systems that can process information and facilitate a response remain the only ground truth for systems that display generalized intelligence capabilities.[9] This perspective aims to explore and juxtapose two broad biomaterial-based approaches toward harnessing neural biological capabilities *in vitro*.

THE CASE FOR *IN VITRO* NEURAL SYSTEMS DEMONSTRATING INFORMATION PROCESSING AND LEARNING

Biological brains display highly efficient learning, both in terms of power consumption and data requirements,[10] and naturally serve as the core inspiration for leveraging neural cells for information processing and even intelligent purposes. The potential to harness the underlying material of biological brains, neural cells, is a promising idea.[9,11] This promise has led to growing research in the development of intelligent information processing systems which integrate biological—typically neural—tissue with synthetic computing systems.[12] Typically, this research uses microelectrode (sometimes called multielectrode) arrays (MEAs).[13] MEAs are devices designed to record and stimulate changes in extracellular electrical activity.[14] Early research using MEAs revealed that *in vitro* neural systems exhibit remarkable adaptability to electrical stimulation.[15–17] Some studies even explored using neural activity to control robotic platforms.[18,19] Later research demonstrated that closed-loop electrophysiological stimulation and recording, even when not tightly coupled with real-time performance, can induce measurable changes in network plasticity.[19,20] Closed-loop electrophysiological interactions in this context involve being able to provide information into a neural culture through electrical stimulation, recording the response from the neural cultures, then using that response to update future electrical stimulations in a rules-based manner. The fundamental importance of closed-loop systems is the ability to provide feedback to neural systems about the outcome of the system's actions. Without this environmental information, learning is logically implausible as learning requires adjusting from some form of feedback following an action.

For example, the impact of a real-time closed-loop system can be seen in an experiment that allowed monolayers of cortical cells to successfully demonstrate basic learning in simulated game-worlds in a rapid time course, showing improvement over time in a simplified Pong game.[21,22] In this study, the real-time closed-loop system took activity from counterbalanced regions across an MEA, applied this to move a stimulated paddle either up or down based on the activity, then stimulated a separate region of the MEA with a pattern that represented the "ball's" relative position to the "paddle," continually updating it as the paddle was moved based on the cultures electrical activity. While in this study a second layer of feedback was applied to the cells based on whether the paddle intercepted the ball or not, this explicit signal is not required for a system to be considered closed loop (although it may alter the outcome from the feedback); simply having the future external information be shaped in some way by the internal activity is the minimum that is required.[23] Of course, not all approaches to explore information processing in neural cells require a closed-loop system. Other studies have adopted a reservoir computing-type approach, where the cell culture is treated as a method of applying non-linear transformations to an input signal to allow better classification of such inputted information.[24] In short, a reservoir computing approach may provide an open-loop pattern of stimulation to the neural cultures, record the response, then see whether that response can be used to understand or classify the original signal with higher fidelity than the input data. These reservoir computing approaches have shown that neural systems may be able to respond to complex stimulations, such as blind source separation and nonlinear equation prediction.[25–28]

TECHNOLOGICAL IMPROVEMENTS PROMISE GREAT FUNCTIONALITY IN THE FUTURE

Developments in exploring how neural systems process information and what level of intelligence they can exhibit have also been aided by more scalable and controllable methods to obtain neural cells. Earlier work typically used neural cells from primary rodent cultures, which, while effective in generating *in vitro* neural tissue, had limited specificity, were not scalable, and had very restricted longevity.[29] A solution to this is to use synthetic biology methods, which have been defined as the application of an engineering paradigm of systems design to biological systems to produce predictable and robust systems.[30] Applying synthetic biology techniques to pluripotent stem cells has permitted highly controllable production of *in vitro* neural tissue through either ontogenetic differentiation or direct overexpression of key genes.[31–33] The generation and refinement of human induced pluripotent stem cell (hiPSC) technology has allowed the starting cellular material to be scalable and ethically produced, while capturing diverse genetic lineages.[34] A variety of terms have already been proposed to describe this area, including biointelligence, neurocomputing, or more specifically, synthetic biological intelligence (SBI).[22,35] SBI has previously been defined as intentionally synthesizing a combination of biological and silicon

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substrates *in vitro* for the purpose of goal-directed or other intelligent behavior.[9] This definition is by intention broad to encompass the multiple directions that the research may progress.

Indeed, despite the rapidly growing body of research that fits the SBI definition, the development of using synthetic biological methods to generate intelligent devices capable of information processing currently is in the early stages.[36–39] Nonetheless, despite the novelty of these approaches, efforts have already commenced to produce bespoke hardware and software to support this technology.[40–42] An important note that unifies these approaches is the acknowledgement that the key function of neural tissue is not just to express electrophysiological activity or biophysical structures but to process and respond to real-time, ideally closed-loop, structured information landscapes. To facilitate this, two divergent—albeit complementary—pathways forward currently exist along a spectrum. The distinction ultimately arises from the conceptual goal of the work. On one end of the spectrum, the focus is to capture physiologically relevant traits of the organ of interest: essentially aiming to leverage a brain-on-a-chip. This focus is perhaps best exemplified by the rise of organoid intelligence (OI) and has attracted significant interest and investment both from the public and the scientific community.[11,26,43–45] OI has been described as an emerging multidisciplinary field working to develop biological computing using three-dimensional (3D) cultures of brain cells.[45] Formally OI has been defined as the aim "to harness the innate biological capabilities of brain organoids for biocomputing and synthetic intelligence by interfacing them with computer technology."[44] Early work apropos of OI has focused on presenting patterned electrophysiological information to organoids and analyzing the responses.[26,46] For instance, in one study a 2D Henon map was decomposed into a 1D spatiotemporal sequence that was input into neural organoids, where the resulting cell activity was able to classify the next step of the sequence significantly better than through linear regression alone.[26] Examples of closed-loop systems have also arisen, where the information inputted to the organoids has been dynamically adjusted based on the response of the organoids.[42,47] One such approach explored the well-established machine learning task "Cartpole," finding that altering the closed-loop stimulation paradigm led to significantly improved performance over time.[47]

Yet, an alternative approach that aims for—and requires—little to no physiological relevance exists. This is best described as a bioengineering approach to intelligent devices: here proposed to be termed "bioengineered intelligence" (BI). Here, the focus would be to leverage the computational complexity of networks of neurons that may hold at least third-order complexity.[48] Under this framework, neural tissue is seen as a material with particular properties where key features are purpose-built. As this perspective is the first to propose the BI concept as a distinct approach, explicit approaches targeting BI are limited at this time; however, work adopting a bottom-up neuroscience perspective would accord with BI.[49,50] This work has explored how to create highly reproducible activity patterns from specified types of electrophysiological stimulation in a controllable and reproducible manner, using methods to control cell placement and connectivity.[49] This approach has also been applied to demonstrate that modular assemblies of neural cells can classify spoken digits encoded into electrical pulses via an MEA.[27] BI would build on these approaches by integrating closed-loop environments, to facilitate more advanced performance within these networks. Formally, BI would be defined as: the use of defined circuits of neural or other cell types assembled in controllable structures for the purpose of information processing, intelligence, and biocomputing. Conceptually, the distinction between OI and BI for how to harness intelligence from neural cells could be considered as a top-down (OI is inspired by *in vivo* brains) vs. bottom-up (BI is inspired by collections of cells processing information in circuits) approach. Yet, conversely, the assembly of these systems would be conceptualized inversely, with BI being a top-down process where macrostructure is determined through biomaterials and OI is a bottom-up process where organoids develop through spontaneous biological processes. However, the differences and implications go further as shown in Figure 1, which provides a framework for conceptualizing different approaches. Figure 2 outlines a roadmap for how to consider these two technologies and how to best utilize the foundational technologies being developed. Finally, the key promises, and potential pitfalls, of each approach are also highlighted along with what an eventual synthesis of these technologies may involve.

THE ORGAN OF INTELLIGENCE INSPIRES OI

The idea of brains, separate from a body, that can display intelligence, even interact with the external world, predates even the earliest modern computers.[51] Neural organoids first arose as an *in vitro* method to study neurodevelopment, particularly differences in functional layering in the cortex.[52–54] Neural organoids are cellular structures formed from 3D aggregates of typically heterogeneous neural cells differentiated from pluripotent stem cells. For example, cortical organoids would aim to contain cell types found in the frontal cortex[32] while a sensorimotor organoid would display cell types found in neuromuscular junctions.[55] The ability to generate a diverse array of neural organoids rapidly facilitated use for *in vitro* drug screening, disease modeling, and studying differences in neural development between species.[55–57] Neural organoids typically not only display more complex morphology than monolayer cultures but also more complex electrophysiological dynamics.[32,58,59] These organoid cultures can also be extended when one organoid can interface with another *in vitro* to form a linked cell culture of organoids (called an assembloid).[60,61] These advancements have not continued without challenges. Interior hypoxia and cell death results in a necrotic core that disrupt continuous neuronal production and limits the development of cortical lamination.[62–65] Neural organoids also show impaired cell-type specification linked to activated cellular stress pathways.[66] Challenges with creating and stabilizing the right microenvironment for long-term organoid culturing also exist.[67] As advancements in organoid technology continue, including the development of vascularized models, which may aid in perfusion under the right conditions,[55,68,69] it is possible these challenges may be resolved.

The potential to harness the capabilities facilitated by *in vivo* brains in these complex 3D cell cultures has spurred research aiming to exploit organoid functionality for intelligent or other

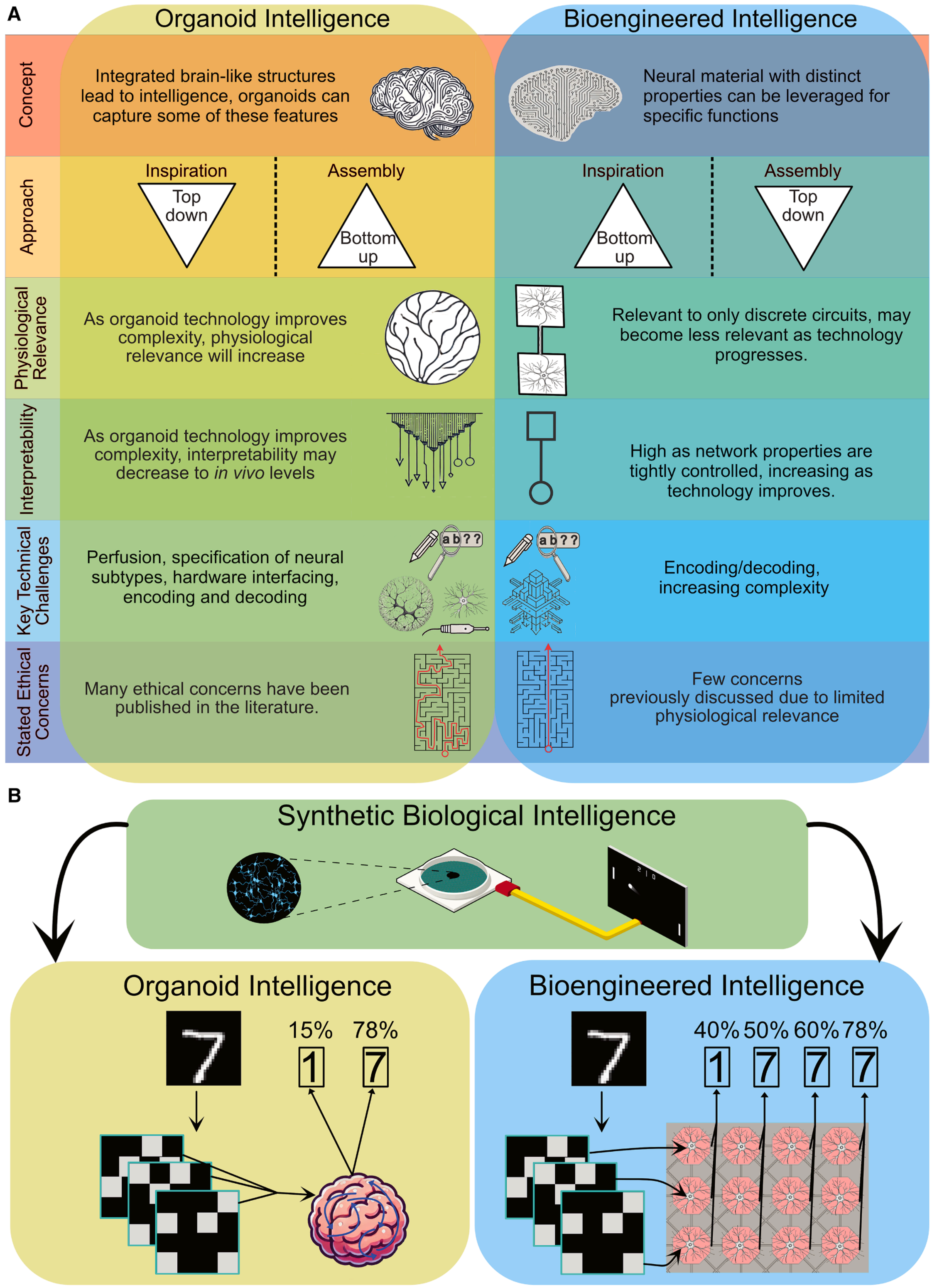

Figure 1. While both OI and BI are examples of SBI, key conceptual differences and algorithms may exist

(A) Key proposed differences for OI and BI pathways.

(B) OI and BI may intersect or diverge, but both could be considered as stemming from general attempts to generate intelligent devices from synthetic biology approach. In this way the earlier term SBI acts as a useful catchall term for the field, while OI and BI would designate distinct approaches. Many experimental

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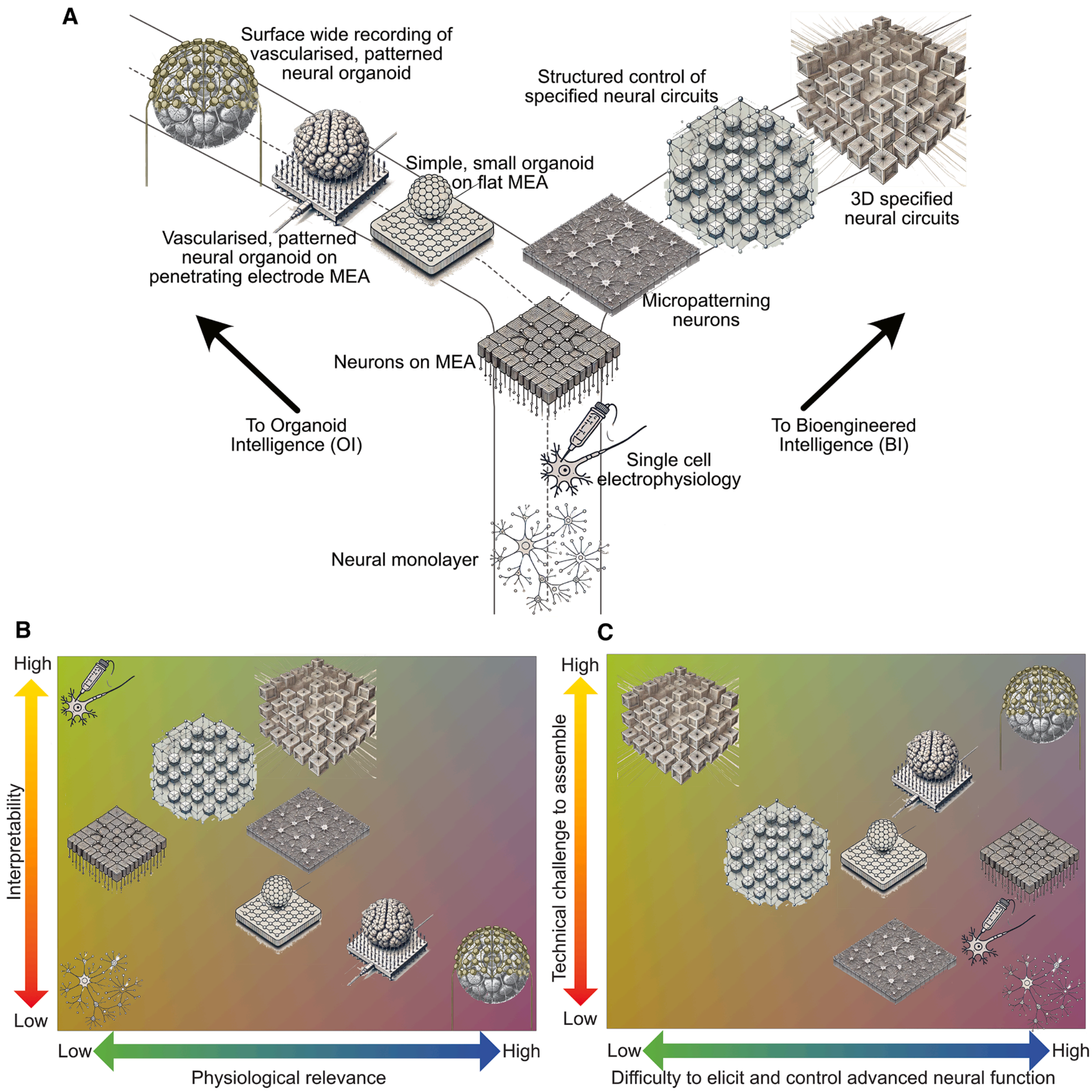

Figure 2. Proposal of possible differential developments of OI and BI pathways, along with where on a series of spectrums each may exist
(A) Schematic showing key concepts of progressing down either an OI or BI pathway with labels for each setup. While eventually these pathways could potentially eventually converge for some uses, there is also the potential that these diverging pathways could remain application specific, each with their own challenges and opportunities.
(B) Uses labels shown in (A). Each system will have varying levels of interpretability for the data generated along with differing degrees of physiological relevance.
(C) Uses labels shown in (A). The technical challenge to assemble these systems will also vary, as will the (assumed) difficulty of reliably eliciting and controlling advanced neural functions for various applications.

information processing tasks.[11,26,45] The approach has specific promises despite the challenges in cultivating the 3D cell cultures outlined above. *In vivo* neurophysiology acts as a ground truth for functioning systems to display intelligent behavior. By recapitulating brain physiology, it is reasonably likely that advanced processes could eventually be performed by these *in vitro* biological simulacra. A key goal of these organoids is for the cultures to display comparable patterning and connectivity to that found *in vivo*. If achieved, work in organoid systems would better reflect human biology and may more reliably inform certain aspects of disease and corresponding interventions than less physiologically relevant *in vitro* structures.[11] Organoids with

setups could be applied to either OI or BI, here some key conceptual differences are shown in a simple open-loop digit recognition example. Here, OI involves feeding in encoded signals and allowing non-linear transformations in activity to result in a collection of signals. A classifier might work on the entire network and result in a confidence score of key options. A similar approach might be adopted for BI; however, different layers of the neural system may be utilized in pre-designed distinct manners, which could result in layers distal from the signal providing maximum accuracy.

this level of physiological relevance and complexity could then be leveraged as information processing systems even if the internal dynamics are obscured (a "black box" approach). This ameliorates the need to understand (presumably) intricate neuroinformatic dynamics that drive these information processing and/or intelligence functions. The black box approach has already been successfully used through reservoir computing-type approaches with varying degrees of nuance and cell culture complexity.[26,27]

Reservoir computing is attractive where internal dynamics are obscured, as it is theoretically simple, requires neither specific feedback to the neural systems nor a deep understanding of the neural system structures.[24] However, the convenience of a reservoir computing approach when interacting with organoids for intelligence also belies the theoretical and technical challenges in meaningfully progressing beyond these open-loop systems. One key issue is that significant variability between individual organoids still exists.[32,57,69] Closed-loop electrophysiological interactions have so far focused primarily on monolayer systems.[19,20,22] In these simpler systems, the relatively minimal structure makes it rational to treat any area of the culture much like any other, assuming a homogeneous distribution of cell types and connections to facilitate flexible place coding of information. In contrast, organoids display meaningful structure and a spatial organization of cell types.[61,63] To further complicate matters, mapping of organoid structures—while maintaining viable function, functional connectivity, and different cell types—is difficult and temporally variable.[59,70] This leads to difficulty in specifying and testing what constitutes a functional unit within an organoid.

The difficulty interacting with neural organoids at discrete functional units is only compounded as most MEA systems are two-dimensional (2D) and thus can only interface with the base surface of an organoid. This concern persists regardless of whether the MEAs are passive with typically 8–256 electrodes for recording, or active high density complementary metal-oxide-semiconductor (CMOS) MEA with 1,024–33,840 routable active electrodes.[71–73] To mitigate this concern, significant progress in designing and building 3D MEA systems has occurred in recent years.[74] Some 3D MEA systems involve structured electrodes, which may partially penetrate into the organoid.[75,76] Given that some of these systems can be based on high-density CMOS MEA, this can be a useful approach to record activity from a high number of sites not immediately on the MEA surface.[77,78] While a detailed discussion of MEA technology and developments is beyond the scope of this perspective, these topics have been covered in recent review articles.[12,29,79] Regardless, even a high number of partially penetrating electrodes can still only facilitate recordings from a relatively isolated plane of the organoids near the adherent base surface as the length of these electrodes do not vary. Therefore, to differing extents, these limits persist for both 2D and 3D MEA systems regardless of density. Two considerations can arise from recognizing this limitation: that interacting with more regions of organoids may be less impactful than some believe, or that more advanced MEA systems are required to overcome this limitation. It can be argued that given a suitable understanding of neural function, the impact of this limitation might be minimal. The first line of reasoning can take inspiration from *in vivo* human neuroanatomy. There exists a relatively minimal laminar region in the brain where all brain/body interactions occur. If the goal of the interaction between neural organoids and the MEA is to facilitate embodiment,[9] then it is reasonable to propose that with sufficient understanding of the neural system a limited plane of reading near the base surface of the neural organoids may likewise be sufficient. With this approach, it would become necessary to be able to infer key dynamic states in the unobserved areas based on the observable system, however, this is a plausible approach.[80] The result would allow greater functional integration of the neural tissue with the MEA than might otherwise be expected from a relatively minimal interaction layer.

The contrary consideration would suggest that interacting with more of the organoid is required, with two broad approaches being explored currently. One case aims to develop systems with longer and even varied penetrating electrodes, although these are not high density.[81–83] Yet, so far, evidence has not demonstrated that penetrating electrodes offer a clear advantage to better characterize functional units in an organoid over 2D MEA. One key concern with penetrating electrodes remains that since the issue of internal necrosis with organoids has not yet been resolved,[62–65] electrophysiological interactions with cells within the organoid core may not be particularly meaningful. This stands as an important area of future research and development. The other approach aims to avoid concerns of the necrotic core and target more of the organoid surface than just the base. Despite MEA systems also currently being restricted to lower density electrodes, this approach is promising. A supporting driver for why recording from more of the surface may be preferable is the wealth of information available simply from the surface of the mammalian cortex.[84] Stretchable mesh nanoelectronics-based systems that do not completely encapsulate the organoid, but do demonstrate minimally invasive recordings from a greater surface of neural organoids than only the base surface, is one approach.[85] Another approach is 3D self-rolled biosensors that are capable of encapsulating organoids and have been found to take viable recordings of cardiac organoids from around the entire surface and have been adapted to neural organoids.[86] Likewise, shell-based, foldable MEAs have also been developed and shown to be capable of stable readings from multiple areas around neural organoid surfaces.[87,88] While these encapsulating systems still require significant refinement and increased electrode density, including improvements to the usability, manufacturability, and access, they do offer viable pathways to capture more widespread surface-based activity from cortical organoids. With appropriate hardware and software to then support these advanced MEA technologies—and assuming the synthetic biology challenges are overcome in developing more advanced organoids—it would then be possible to interact with neural organoids in highly sophisticated ways beyond our current capabilities.[45]

SYNTHETIC BIOLOGY BUILDING BLOCKS FOR BI

Compared with the use of organoids as a base for intelligent information processing devices as OI,[11,26,43–45] a unified approach of building and designing biological neural circuits from base principles at scale has been minimal. This is particularly surprising when considering that the field of neuromorphic chip design is largely inspired by the aim of essentially capturing these dynamics in hardware.[48,89,90] However, even simple biological neural networks are capable of displaying the underlying properties required for more complex functions, specifically network plasticity and nonlinear computing capacity.[37] Therefore, these core capabilities may theoretically be leveraged for highly sophisticated applications. BI is proposed here as a term to designate this technology and distinguish it from OI. Unlike OI, which aims to capture physiologically relevant features, BI can take a more utility-focused approach in line with calls for a bottom-up neuroscience approach.[91] These systems can be reconfigured to suit key purposes, leveraging the entire synthetic biological toolkit and even adopting an evolutionary design approach, as previously proposed for non-neural cell types.[92] However, unlike the organoid approach, where the aim is to create a biological neural system physiologically comparable to a functioning organism, there is no ground truth of intelligent function. While specific brain circuits have been linked to certain functions in living organisms, little to no reproducible evidence has yet been gathered that recreating any specific circuit in isolation will yield similar complex functionality—although it is a compelling direction of ongoing and future research.

Despite the uncertainty as to what may be achieved, multiple independent converging lines of innovative research have established that even simple neural circuits display key functional properties that would facilitate at least core information processing and even intelligent functions. Even unorganized monolayers of neurons robustly showcase neural plasticity and responsiveness to input.[15,16,20,93] Neural monolayer cultures also display population-wide coherent activity through an interplay between network dynamics and topology via a noise-focusing effect.[94,95] Moreover, just because an organoid approach is not adopted, does not mean that this bioengineering approach is restricted to unorganized monolayers. The use of 3D scaffold-based micropatterning has enabled more complex *in vitro* cortical networks that show patterning akin to cortical lamination.[96] Interestingly, when cultured on 3D scaffolds, as opposed to monolayers, the functional connectivity of these networks demonstrated more modular microcircuits, highlighting that neural circuit connectivity is modifiable through the physical environment.[96] To create a physical environment to control the topology and geometry of *in vitro* neural network circuitry, two main approaches are typically used: either micropatterned surfaces or structures that physically constrain neurons.[91]

Micropatterning involves coating the surface on which cells are cultured with specific molecules.[97] Envisioned as a method to position cells distinctly on a circuit or to engineer specific structures, micropatterning with an extracellular matrix (ECM) permitted control of neural cell adhesion, proliferation, differentiation, and neurite outgrowth.[98] Examination of synaptic plasticity and neural connectivity patterns found these traits to be maintained despite micropatterning.[99] Lithography-based approaches have also been used to control cell adhesion, neurite outgrowth, and connectivity with a high degree of specificity.[100,101] Micropatterning will be particularly useful for groups that lack the facilities to generate structures that physically constrain neurons. Micropatterning can also allow for rapid and cost-effective iterations on neural cell arrangement. In this way, micropatterning can also be more flexibly combined with alternative approaches. However, micron scale patterning can degrade rapidly, and although some report that larger patterning may persist for at least several months, most report about 5 weeks.[97]

The alternative approach—using structures that physically constrain neurons—is capable of supporting neural structures for significantly longer periods of time than through micropatterning alone.[102] These structures, sometimes referred to as microfluidics, are typically generated from polydimethylsiloxane (PDMS).[102] PDMS devices can be constructed with an enormous variety of topology and geometry. *In vitro* research using similar structures shows that network formation can be controlled by limiting where neurons can adhere to treated surfaces, along with directing connective structures like axon and dendrites through size-constrained microchannels.[103,104] It is also possible to use barbed channels to create unidirectional connectivity between distal neurons separated into different regions.[105,106] Intricate patterns can be produced in PDMS structures and combined with hiPSCs to build small and controllable circuits.[49,50,91] By integrating these PDMS structures onto MEA, electrophysiological stimulation and recording can occur in specified ways.[105,107] Of note, introducing a modular architecture via the PDMS structures to neural networks enriches spontaneous activity compared with unstructured networks.[108–110] Controlled neural networks can even be entrained to showcase a consistent stimulation-induced electrophysiological response.[49,50] Given these dynamic features, it is coherent that research has also successfully showcased that using neural networks with modular architecture can also act in a reservoir computing manner for spoken digit classification where increased modularity of the neural networks was correlated with increased accuracy.[27] This is consistent with other work that has shown that modular architectures can increase the sensitivity of the neural network to external stimulation and increase plasticity in the neuronal ensembles.[111,112]

Theoretically these approaches of directing and controlling the neural networks could create more specific computational paradigms, reducing biological noise and improving outcomes. Computational models exploring the impact of combining different cell types or creating discrete circuitry supports the idea that noise can be reduced and/or managed to improve accuracy in processing.[113,114] For a BI system not focused on modeling *in vivo* neural physiology, it would be acceptable to construct these systems to achieve a desired outcome even if it is not physiologically relevant at the macro-scale. The generation of specific cell types through synthetic biology methods can provide even greater control over these circuits.[115] Overall, the successful integration of an approach harnessing the underlying fundamental principles of neural intelligence would accord well with a collective intelligence framework, providing a data point for what different systems of different complexity can achieve.[116] While this approach will likely require a more refined understanding of how to drive these neural systems toward desired outcomes, it will in turn provide a more controllable system to gain these understandings through empirical investigations. The ultimate outcome of a BI approach may be an intelligent system with only a vague resemblance to current mammalian brains, where cells are treated and used as discrete materials with well-understood—albeit remarkable—properties.

OPPORTUNITIES AND CHALLENGES

Pathways for both OI and BI offer unique challenges and opportunities. While it is possible to imagine combinations of these approaches, the different conceptual groundings and sufficient challenges in either approach alone make it useful to consider each independently. For instance, the practical applications of these two pathways may differ. Currently, both OI and BI can be used in basic and translational research as demonstrated by early related work,[46,117] yet likely with different focuses. For instance, OI will be useful in understanding how information processing may change during neurodevelopment or following the administration of pharmaceuticals that differentially permeate neural tissue. In contrast, BI is a more appropriate pathway to understand how specific circuits may manage information and respond to specific patterns of stimulation. In the short term, OI may be useful to capture basic information processing functions as seen in basic organisms, modeling key physiological features of the organ of interest during information processing and more accurately modeling disease and treatment responses. In contrast, BI may not follow these physiological inspirations, instead focusing on how to pass a pattern of information through a defined circuit to optimize for a particular outcome (an extension on what Figure 1B illustrates). It might also be argued that BI would be more amenable to greater genetic modification of the underlying substrates to unlock capabilities not possible otherwise, given the reduced emphasis on physiological relevance. Yet, similar approaches may be equally viable for OI approaches.

Long-term predictionsx become increasingly difficult and the feasibility of knowing where this work could progress is limited. One could propose that the eventual success of OI could lead to organism-level information processing and intelligent functions. Yet, whether this complexity would mirror that of an insect, canine, or even a human cannot be known at this point. Such an advanced system would have clear applications in robotics, as navigating complex and dynamic environments is something in which biological brains excel. The end goal of BI is even more opaque, as the limitations and advantages for this approach cannot be accurately judged based on existing living neural systems. It is plausible that advanced implementations of BI will end up incredibly similar to OI, adopting a modular—albeit 3D—geometry if current physiological organizations are optimal and inflexible. If so, then the applications would be closely interrelated and focus on what is currently known about what neural systems perform optimally.[118] However, it is also possible that new geometric and topological organizations will be discovered that provide enhanced information processing and intelligent functions that go beyond what is currently known physiologically. It must be emphasized that these potential applications are only speculative, proposed here with the intention of stimulating further consideration and exploration. Additionally, a key future direction will involve exploring how to integrate these approaches, interfacing organoids with discrete structures, or building modular controllable networks of organoids.

Naturally, discussions of future applications also raise ethical considerations, where there is value in considering these pathways distinctly. More concerns have been raised about organoids giving rise to morally relevant states than discrete BI systems.[119–122] The core arguments for ethical concerns with brain organoids focus on the attempts to build physiologically comparable neural structures to animals and even humans.[123] Some have even discussed whether a precautionary principle framework is most appropriate to manage these concerns.[124,125] Yet, it is generally accepted in the scientific community that these concerns can currently be addressed through less prohibitive frameworks.[45,119,126–129] Nonetheless, it is still generally recognized that increasing the complexity of self-contained neural systems beyond a certain point may require additional neuroethical frameworks. A tightly managed anticipatory governance approach has been proposed as the optimal way forward.[9,130] In contrast, BI approaches appear to be perceived as less ethically fraught as they have received relatively minimal attention ethically; however, the reduced concern may rely on the assumption that these morally relevant states are dependent on a specific physiologically relevant structure, an assumption that may or may not be true. If that assumption is challenged, the implications would be considerable, requiring a detailed examination of what states other complex systems—such as AI—may possess and how they would relate to biological substrates.[131] As BI capabilities expand, it is likely that more ethical questions will arise but being distal from (at least mammalian) physiological relevance may greatly ameliorate concerns about the risk that purpose-built systems could possess morally relevant states. Functionally, the drive for OI to capture physiologically relevant forms and functions provides a greater likelihood of complex function than the more bottom-up BI approach, but it may restrict the nuance that can be leveraged from the system.

CONCLUSIONS: THE NEED TO TRAVEL BOTH ROADS

Overall, this perspective has aimed to contrast the rapidly growing field of OI with the complementary but distinct approach of BI proposed here. The key intention is to demonstrate that multiple pathways exist to explore how to best leverage *in vitro* neural systems for information processing and intelligent applications. While many nuances between the two approaches exist, the appropriate approach will ultimately depend on the purpose and intention of the research question or eventual application. OI at its core aims to recreate the brain, or at least a unit thereof, as a model of an organ for information processing, while BI uses individual networks in ways not bound by physiological relevance. Naturally, these differences exist on a spectrum. For instance, work on neural assembloids has allowed some functional modularity of the organoid subcomponents to be controlled.[61,132–134] In these works, the goal is primarily to target physiological relevance, so these approaches would fall closer to OI than BI, yet variations could change where on this spectrum future work may fall. In contrast, work that examines functional circuit formation may be interested in physiologically relevant mechanisms, yet if it takes a reductive bottom-up approach, it is better described as closer to a BI approach if the resulting neural culture were used for information processing or intelligence purposes.[135] While such approaches may blur the lines of the framework proposed in this article, by making the two approaches explicit, it will hopefully allow better communication of future research and development efforts in an appropriate way, allowing distinct approaches and goals to be appreciated for their unique merits. This does not prevent these methods being combined or for additional subspecializations in the field of SBI to be proposed. Regardless of the potential of either approach, for the purposes of scientific discovery, juxtaposing results across scales of complexity will yield insights not possible with any single method and is proposed here as a key future direction for the field.[126] Therefore, ultimately, while it is incredibly likely that in the future these two paths will in some cases converge and in others further diverge in nuance, it is a synthesis of the approaches that will provide the greatest yield to scientific progress.

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AUTHOR CONTRIBUTIONS

B.J.K. conceived and wrote the manuscript.

DECLARATION OF INTERESTS

B.J.K. is an employee of and holds shares in Cortical Labs, a start-up exploring work related to this space. B.J.K. holds an interest in patents related to this space.

REFERENCES

1. Eloundou, T., Manning, S., Mishkin, P., and Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. Science *384*, 1306–1308. https://doi.org/10.1126/science.adj0998.
2. Peng, Y., Han, J., Zhang, Z., Fan, L., Liu, T., Qi, S., Feng, X., Ma, Y., Wang, Y., and Zhu, S.-C. (2024). The Tong Test: Evaluating Artificial General Intelligence Through Dynamic Embodied Physical and Social Interactions. Engineering *34*, 12–22. https://doi.org/10.1016/j.eng.2023.07.006.
3. Qu, Y., Wei, C., Du, P., Che, W., Zhang, C., Ouyang, W., Bian, Y., Xu, F., Hu, B., Du, K., et al. (2024). Integration of cognitive tasks into artificial general intelligence test for large models. iScience *27*, 109550. https://doi.org/10.1016/j.isci.2024.109550.
4. Kagan, B.J., Mahlis, M., Bhat, A., Bongard, J., Cole, V.M., Corlett, P., Gyngell, C., Hartung, T., Jupp, B., Levin, M., et al. (2024). Toward a nomenclature consensus for diverse intelligent systems: Call for collaboration. Innovation *5*, 100658. https://doi.org/10.1016/j.xinn.2024.100658.
5. Kagan, B.J., Razi, A., Bhat, A., Kitchen, A.C., Tran, N.T., Habibollahi, F., Khajehnejad, M., Parker, B.J., Rollo, B., and Friston, K.J. (2023). Scientific communication and the semantics of sentience. Neuron *111*, 606–607. https://doi.org/10.1016/j.neuron.2023.02.008.
6. Gignac, G.E., and Szodorai, E.T. (2024). Defining intelligence: Bridging the gap between human and artificial perspectives. Intelligence *104*, 101832. https://doi.org/10.1016/j.intell.2024.101832.
7. Klein, C., and Barron, A.B. (2024). Comparing cognition across major transitions using the hierarchy of formal automata. Wiley Interdiscip. Rev. Cogn. Sci. *15*, e1680. https://doi.org/10.1002/wcs.1680.
8. Yenduri, G., Ramalingam, M., Selvi, G.C., Supriya, Y., Srivastava, G., Maddikunta, P.K.R., Raj, G.D., Jhaveri, R.H., Prabadevi, B., Wang, W., et al. (2024). GPT (Generative Pre-Trained Transformer)— A Comprehensive Review on Enabling Technologies, Potential Applications, Emerging Challenges, and Future Directions. IEEE Access *12*, 54608–54649. https://doi.org/10.1109/ACCESS.2024.3389497.
9. Kagan, B.J., Gyngell, C., Lysaght, T., Cole, V.M., Sawai, T., and Savulescu, J. (2023). The technology, opportunities, and challenges of Synthetic Biological Intelligence. Biotechnol. Adv. *68*, 108233. https://doi.org/10.1016/j.biotechadv.2023.108233.
10. Lansner, A., Fiebig, F., and Herman, P. (2023). Fast Hebbian plasticity and working memory. Curr. Opin. Neurobiol. *83*, 102809. https://doi.org/10.1016/j.conb.2023.102809.
11. Smirnova, L. (2024). Biocomputing with organoid intelligence. Nat. Rev. Bioeng. *2*, 633–634. https://doi.org/10.1038/s44222-024-00200-6.
12. Kagan, B.J., Habibollahi, F., Watmuff, B., Azadi, A., Doensen, F., Loeffler, A., Byun, S.H., Servais, B., Desouza, C., Abu-Bonsrah, K.D., et al. (2025). Harnessing intelligence from brain cells in vitro. Neuroscientist, 10738584251321438. https://doi.org/10.1177/10738584251321438.
13. Mierzejewski, M., Steins, H., Kshirsagar, P., and Jones, P.D. (2020). The noise and impedance of microelectrodes. J. Neural Eng. *17*, 052001. https://doi.org/10.1088/1741-2552/abb3b4.
14. Middya, S., Curto, V.F., Fernández-Villegas, A., Robbins, M., Gurke, J., Moonen, E.J.M., Kaminski Schierle, G.S., and Malliaras, G.G. (2021). Microelectrode Arrays for Simultaneous Electrophysiology and Advanced Optical Microscopy. Adv. Sci. (Weinh) *8*, 2004434. https://doi.org/10.1002/advs.202004434.
15. Shahaf, G., and Marom, S. (2001). Learning in Networks of Cortical Neurons. J. Neurosci. *21*, 8782–8788. https://doi.org/10.1523/JNEUROSCI.21-22-08782.2001.
16. Jimbo, Y., Robinson, H.P.C., and Kawana, A. (1998). Strengthening of synchronized activity by tetanic stimulation in cortical cultures: application of planar electrode arrays. IEEE Trans. Bio Med. Eng. *45*, 1297–1304. https://doi.org/10.1109/10.725326.
17. Pasquale, V., Massobrio, P., Bologna, L.L., Chiappalone, M., and Martinoia, S. (2008). Self-organization and neuronal avalanches in networks of dissociated cortical neurons. Neuroscience *153*, 1354–1369. https://doi.org/10.1016/j.neuroscience.2008.03.050.
18. Chao, Z.C., Bakkum, D.J., Wagenaar, D.A., and Potter, S.M. (2005). Effects of random external background stimulation on network synaptic stability after tetanization: A modeling study. Neuroinformatics *3*, 263–280. https://doi.org/10.1385/NI:3:3:263.
19. Tessadori, J., Bisio, M., Martinoia, S., and Chiappalone, M. (2012). Modular Neuronal Assemblies Embodied in a Closed-Loop Environment: Toward Future Integration of Brains and Machines. Front. Neural Circuits *6*, 99. https://doi.org/10.3389/fncir.2012.00099.
20. Bakkum, D.J., Chao, Z.C., and Potter, S.M. (2008). Spatio-temporal electrical stimuli shape behavior of an embodied cortical network in a goal-directed learning task. J. Neural Eng. *5*, 310–323. https://doi.org/10.1088/1741-2560/5/3/004.
21. Habibollahi, F., Kagan, B.J., Burkitt, A.N., and French, C. (2023). Critical dynamics arise during structured information presentation within embodied in vitro neuronal networks. Nat. Commun. *14*, 5287. https://doi.org/10.1038/s41467-023-41020-3.
22. Kagan, B.J., Kitchen, A.C., Tran, N.T., Habibollahi, F., Khajehnejad, M., Parker, B.J., Bhat, A., Rollo, B., Razi, A., and Friston, K.J. (2022). In vitro neurons learn and exhibit sentience when embodied in a simulated game-world. Neuron *110*, 3952–3969.e8. https://doi.org/10.1016/j.neuron.2022.09.001.

23. Friston, K.J., Salvatori, T., Isomura, T., Tschantz, A., Kiefer, A., Verbelen, T., Koudahl, M., Paul, A., Parr, T., Razi, A., et al. (2025). Active Inference and Intentional Behaviour. Preprint at arXiv. Neural Comput. *37*, 666–700. https://doi.org/10.1162/neco_a_01738.

24. Schöfmann, C.M., Fasli, M., and Barros, M.T. (2024). Investigating Biologically Plausible Neural Networks for Reservoir Computing Solutions. IEEE Access *12*, 50698–50709. https://doi.org/10.1109/ACCESS.2024.3385339.

25. Isomura, T., Kotani, K., and Jimbo, Y. (2015). Cultured Cortical Neurons Can Perform Blind Source Separation According to the Free-Energy Principle. PLoS Comput. Biol. *11*, e1004643. https://doi.org/10.1371/journal.pcbi.1004643.

26. Cai, H., Ao, Z., Tian, C., Wu, Z., Liu, H., Tchieu, J., Gu, M., Mackie, K., and Guo, F. (2023). Brain organoid reservoir computing for artificial intelligence. Nat. Electron. *6*, 1032–1039. https://doi.org/10.1038/s41928-023-01069-w.

27. Sumi, T., Yamamoto, H., Katori, Y., Ito, K., Moriya, S., Konno, T., Sato, S., and Hirano-Iwata, A. (2023). Biological neurons act as generalization filters in reservoir computing. Proc. Natl. Acad. Sci. USA *120*, e2217008120. https://doi.org/10.1073/pnas.2217008120.

28. Lindell, T.A.E., Ramstad, O.H., Sandvig, I., Sandvig, A., and Nichele, S. (2024). Chaotic Time Series Prediction in Biological Neural Network Reservoirs on Microelectrode Arrays. In International Joint Conference on Neural Networks (IJCNN) (IEEE), pp. 1–10. https://doi.org/10.1109/IJCNN60899.2024.10650567.

29. Cerina, M., Piastra, M.C., and Frega, M. (2023). The potential of in vitro neuronal networks cultured on micro electrode arrays for biomedical research. Prog. Biomed. Eng. *5*, 032002. https://doi.org/10.1088/2516-1091/acce12.

30. Ho, C., and Morsut, L. (2021). Novel synthetic biology approaches for developmental systems. Stem Cell Rep. *16*, 1051–1064. https://doi.org/10.1016/j.stemcr.2021.04.007.

31. Shan, X., Zhang, A., Rezzonico, M.G., Tsai, M.-C., Sanchez-Priego, C., Zhang, Y., Chen, M.B., Choi, M., Andrade López, J.M.A., Phu, L., et al. (2024). Fully defined NGN2 neuron protocol reveals diverse signatures of neuronal maturation. Cell Rep. Methods *4*, 100858. https://doi.org/10.1016/j.crmeth.2024.100858.

32. Fitzgerald, M.Q., Chu, T., Puppo, F., Blanch, R., Chillón, M., Subramaniam, S., and Muotri, A.R. (2024). Generation of 'semi-guided' cortical organoids with complex neural oscillations. Nat. Protoc. *19*, 2712–2738. https://doi.org/10.1038/s41596-024-00994-0.

33. Hergenreder, E., Minotti, A.P., Zorina, Y., Oberst, P., Zhao, Z., Munguba, H., Calder, E.L., Baggiolini, A., Walsh, R.M., Liston, C., et al. (2024). Combined small-molecule treatment accelerates maturation of human pluripotent stem cell-derived neurons. Nat. Biotechnol. *42*, 1515–1525. https://doi.org/10.1038/s41587-023-02031-z.

34. Buckberry, S., Liu, X., Poppe, D., Tan, J.P., Sun, G., Chen, J., Nguyen, T. V., De Mendoza, A., Pflueger, J., Frazer, T., et al. (2023). Transient naive reprogramming corrects hiPS cells functionally and epigenetically. Nature *620*, 863–872. https://doi.org/10.1038/s41586-023-06424-7.

35. Barros, M.T., Kagan, B.J., Hartung, T., and Smirnova, L. (2024). Editorial: Intersection between the biological and digital: synthetic biological intelligence and organoid intelligence. Front. Cell. Neurosci. *18*, 1542629. https://doi.org/10.3389/fncel.2024.1542629.

36. Goldwag, J., and Wang, G. (2023). DishBrain plays Pong and promises more. Nat. Mach. Intell. *5*, 568–569. https://doi.org/10.1038/s42256-023-00666-w.

37. Chen, Z., Liang, Q., Wei, Z., Chen, X., Shi, Q., Yu, Z., and Sun, T. (2023). An Overview of In Vitro Biological Neural Networks for Robot Intelligence. Cyborg Bionic Syst. *4*, 0001. https://doi.org/10.34133/cbsystems.0001.

38. Tanveer, M.S., Patel, D., Schweiger, H.E., Abu-Bonsrah, K.D., Watmuff, B., Azadi, A., Pryshchep, S., Narayanan, K., Puleo, C., Natarajan, K., et al. (2025). Starting a synthetic biological intelligence lab from scratch. Patterns *6*, 101232. https://doi.org/10.1016/j.patter.2025.101232.

39. Shao, W.-W., Shao, Q., Xu, H.-H., Qiao, G.-J., Wang, R.-X., Ma, Z.-Y., Meng, W.-W., Yang, Z.-B., Zang, Y.-L., and Li, X.-H. (2025). Repetitive training enhances the pattern recognition capability of cultured neural networks. PLoS Comput. Biol. *21*, e1013043. https://doi.org/10.1371/journal.pcbi.1013043.

40. Kagan, B.J. (2025). The CL1 as a platform technology for leveraging biological neural system functions. Nat. Rev. Bioeng. https://doi.org/10.1038/s44222-025-00340-3.

41. Zhang, X., Dou, Z., Kim, S.H., Upadhyay, G., Havert, D., Kang, S., Kazemi, K., Huang, K.Y., Aydin, O., Huang, R., et al. (2024). Mind In Vitro Platforms: Versatile, Scalable, Robust, and Open Solutions to Interfacing with Living Neurons. Adv. Sci. *11*, e2306826. https://doi.org/10.1002/advs.202306826.

42. Jordan, F.D., Kutter, M., Comby, J.-M., Brozzi, F., and Kurtys, E. (2024). Open and remotely accessible Neuroplatform for research in wetware computing. Front. Artif. Intell. *7*, 1376042. https://doi.org/10.3389/frai.2024.1376042.

43. Hartung, T., Smirnova, L., Morales Pantoja, I.E., Akwaboah, A., Alam El Din, D.-M., Berlinicke, C.A., Boyd, J.L., Caffo, B.S., Cappiello, B., Cohen-Karni, T., et al. (2023). The Baltimore declaration toward the exploration of organoid intelligence. Front. Sci. *1*, 1068159. https://doi.org/10.3389/fsci.2023.1068159.

44. Morales Pantoja, I.E., Smirnova, L., Muotri, A.R., Wahlin, K.J., Kahn, J., Boyd, J.L., Gracias, D.H., Harris, T.D., Cohen-Karni, T., Caffo, B.S., et al. (2023). First Organoid Intelligence (OI) workshop to form an OI community. Front. Artif. Intell. *6*, 1116870. https://doi.org/10.3389/frai.2023.1116870.

45. Smirnova, L., Caffo, B.S., Gracias, D.H., Huang, Q., Morales Pantoja, I.E., Tang, B., Zack, D.J., Berlinicke, C.A., Boyd, J.L., Harris, T.D., et al. (2023). Organoid intelligence (OI): the new frontier in biocomputing and intelligence-in-a-dish. Front. Sci. *1*. https://doi.org/10.3389/fsci.2023.1017235.

46. Alam El Din, D.-M., Moenkemoeller, L., Loeffler, A., Habibollahi, F., Schenkman, J., Mitra, A., Van Der Molen, T., Ding, L., Laird, J., Schenke, M., et al. (2024). Human Neural Organoid Microphysiological Systems Show the Building Blocks Necessary for Basic Learning and Memory. Preprint at bioRxiv. https://doi.org/10.1101/2024.09.17.613333.

47. Robbins, A., Schweiger, H.E., Hernandez, S., Spaeth, A., Voitiuk, K., Parks, D.F., Van Der Molen, T., Geng, J., Sharf, T., Mostajo-Radji, M. A., et al. (2024). Goal-Directed Learning in Cortical Organoids. Preprint at bioRxiv. https://doi.org/10.1101/2024.12.07.627350.

48. Kumar, S., Williams, R.S., and Wang, Z. (2020). Third-order nanocircuit elements for neuromorphic engineering. Nature *585*, 518–523. https://doi.org/10.1038/s41586-020-2735-5.

49. Duru, J., Maurer, B., Giles Doran, C., Jelitto, R., Küchler, J., Ihle, S.J., Ruff, T., John, R., Genocchi, B., and Vörös, J. (2023). Investigation of the input-output relationship of engineered neural networks using high-density microelectrode arrays. Biosens. Bioelectron. *239*, 115591. https://doi.org/10.1016/j.bios.2023.115591.

50. Girardin, S., Ihle, S.J., Menghini, A., Krubner, M., Tognola, L., Duru, J., Fruh, I., Müller, M., Ruff, T., and Vörös, J. (2023). Engineering circuits of human iPSC-derived neurons and rat primary glia. Front. Neurosci. *17*, 1103437. https://doi.org/10.3389/fnins.2023.1103437.

51. Bernal, J.D. (1929). The World, the Flesh and the Devil: an Enquiry into the Future of the Three Enemies of the Rational Soul, Second Edition (Verso).

52. Kadoshima, T., Sakaguchi, H., Nakano, T., Soen, M., Ando, S., Eiraku, M., and Sasai, Y. (2013). Self-organization of axial polarity, inside-out layer pattern, and species-specific progenitor dynamics in human ES cell–derived neocortex. Proc. Natl. Acad. Sci. USA *110*, 20284–20289. https://doi.org/10.1073/pnas.1315710110.

53. Lancaster, M.A., Renner, M., Martin, C.-A., Wenzel, D., Bicknell, L.S., Hurles, M.E., Homfray, T., Penninger, J.M., Jackson, A.P., and Knoblich, J.A. (2013). Cerebral organoids model human brain development and microcephaly. Nature *501*, 373–379. https://doi.org/10.1038/nature12517.

54. Lancaster, M.A., and Knoblich, J.A. (2014). Generation of cerebral organoids from human pluripotent stem cells. Nat. Protoc. *9*, 2329–2340. https://doi.org/10.1038/nprot.2014.158.

55. Pereira, J.D., DuBreuil, D.M., Devlin, A.-C., Held, A., Sapir, Y., Berezovski, E., Hawrot, J., Dorfman, K., Chander, V., and Wainger, B.J. (2021). Human sensorimotor organoids derived from healthy and amyotrophic lateral sclerosis stem cells form neuromuscular junctions. Nat. Commun. *12*, 4744. https://doi.org/10.1038/s41467-021-24776-4.

56. Kim, J., Koo, B.-K., and Knoblich, J.A. (2020). Human organoids: model systems for human biology and medicine. Nat. Rev. Mol. Cell Biol. *21*, 571–584. https://doi.org/10.1038/s41580-020-0259-3.

57. Eichmüller, O.L., and Knoblich, J.A. (2022). Human cerebral organoids — a new tool for clinical neurology research. Nat. Rev. Neurol. *18*, 661–680. https://doi.org/10.1038/s41582-022-00723-9.

58. Sakaguchi, H., Ozaki, Y., Ashida, T., Matsubara, T., Oishi, N., Kihara, S., and Takahashi, J. (2019). Self-Organized Synchronous Calcium Transients in a Cultured Human Neural Network Derived from Cerebral Organoids. Stem Cell Rep. *13*, 458–473. https://doi.org/10.1016/j.stemcr.2019.05.029.

59. Osaki, T., Duenki, T., Chow, S.Y.A., Ikegami, Y., Beaubois, R., Levi, T., Nakagawa-Tamagawa, N., Hirano, Y., and Ikeuchi, Y. (2024). Complex activity and short-term plasticity of human cerebral organoids reciprocally connected with axons. Nat. Commun. *15*, 2945. https://doi.org/10.1038/s41467-024-46787-7.

60. Miura, Y., Li, M.-Y., Birey, F., Ikeda, K., Revah, O., Thete, M.V., Park, J.-Y., Puno, A., Lee, S.H., Porteus, M.H., et al. (2020). Generation of human striatal organoids and cortico-striatal assembloids from human pluripotent stem cells. Nat. Biotechnol. *38*, 1421–1430. https://doi.org/10.1038/s41587-020-00763-w.

61. Reumann, D., Krauditsch, C., Novatchkova, M., Sozzi, E., Wong, S.N., Zabolocki, M., Priouret, M., Doleschall, B., Ritzau-Reid, K.I., Piber, M., et al. (2023). In vitro modeling of the human dopaminergic system using spatially arranged ventral midbrain–striatum–cortex assembloids. Nat. Methods *20*, 2034–2047. https://doi.org/10.1038/s41592-023-02080-x.

62. Chiaradia, I., and Lancaster, M.A. (2020). Brain organoids for the study of human neurobiology at the interface of in vitro and in vivo. Nat. Neurosci. *23*, 1496–1508. https://doi.org/10.1038/s41593-020-00730-3.

63. Qian, X., Su, Y., Adam, C.D., Deutschmann, A.U., Pather, S.R., Goldberg, E.M., Su, K., Li, S., Lu, L., Jacob, F., et al. (2020). Sliced Human Cortical Organoids for Modeling Distinct Cortical Layer Formation. Cell Stem Cell *26*, 766–781.e9. https://doi.org/10.1016/j.stem.2020.02.002.

64. Andrews, M.G., and Kriegstein, A.R. (2022). Challenges of Organoid Research. Annu. Rev. Neurosci. *45*, 23–39. https://doi.org/10.1146/annurev-neuro-111020-090812.

65. Tang, X.-Y., Wu, S., Wang, D., Chu, C., Hong, Y., Tao, M., Hu, H., Xu, M., Guo, X., and Liu, Y. (2022). Human organoids in basic research and clinical applications. Signal Transduct. Target. Ther. *7*, 168. https://doi.org/10.1038/s41392-022-01024-9.

66. Bhaduri, A., Andrews, M.G., Mancia Leon, W., Jung, D., Shin, D., Allen, D., Jung, D., Schmunk, G., Haeussler, M., Salma, J., et al. (2020). Cell stress in cortical organoids impairs molecular subtype specification. Nature *578*, 142–148. https://doi.org/10.1038/s41586-020-1962-0.

67. Luo, D., Xu, J., Liu, F., and Gu, Z. (2024). Advances and Challenges in Cerebral Organoids Research. Adv. Nanobiomed. Res. *4*, 2300126. https://doi.org/10.1002/anbr.202300126.

68. Mansour, A.A., Gonçalves, J.T., Bloyd, C.W., Li, H., Fernandes, S., Quang, D., Johnston, S., Parylak, S.L., Jin, X., and Gage, F.H. (2018). An in vivo model of functional and vascularized human brain organoids. Nat. Biotechnol. *36*, 432–441. https://doi.org/10.1038/nbt.4127.

69. Nwokoye, P.N., and Abilez, O.J. (2024). Bioengineering methods for vascularizing organoids. Cell Rep. Methods *4*, 100779. https://doi.org/10.1016/j.crmeth.2024.100779.

70. Sharf, T., van der Molen, T., Glasauer, S.M.K., Guzman, E., Buccino, A. P., Luna, G., Cheng, Z., Audouard, M., Ranasinghe, K.G., Kudo, K., et al. (2022). Functional neuronal circuitry and oscillatory dynamics in human brain organoids. Nat. Commun. *13*, 4403. https://doi.org/10.1038/s41467-022-32115-4.

71. Kato, Y., Matoba, Y., Honda, K., Ogawa, K., Shimizu, K., Maehara, M., Fujiwara, A., Odawara, A., Yamane, C., Kimizuka, N., et al. (2020). High Density and Large-Scale MEA System Featuring 236,880 Electrodes at 11.72μm Pitch for Neuronal Network Analysis. In IEEE Symposium on VLSI Circuits (IEEE), pp. 1–2. https://doi.org/10.1109/VLSICircuits18222.2020.9162947.

72. O'Leary, G., Khramtsov, I., Ramesh, R., Perez-Ignacio, A., Shah, P., Moradi Chameh, H., Gierlach, A., Genov, R., and Valiante, T.A. (2022). OpenMEA: Open-Source Microelectrode Array Platform for Bioelectronic Interfacing. Preprint at bioRxiv. https://doi.org/10.1101/2022.11.11.516234.

73. Kim, D., Kang, H., and Nam, Y. (2020). Compact 256-channel multi-well microelectrode array system for *in vitro* neuropharmacology test. Lab Chip *20*, 3410–3422. https://doi.org/10.1039/D0LC00384K.

74. Choi, J.S., Lee, H.J., Rajaraman, S., and Kim, D.-H. (2021). Recent advances in three-dimensional microelectrode array technologies for in vitro and in vivo cardiac and neuronal interfaces. Biosens. Bioelectron. *171*, 112687. https://doi.org/10.1016/j.bios.2020.112687.

75. Steins, H., Mierzejewski, M., Brauns, L., Stumpf, A., Kohler, A., Heusel, G., Corna, A., Herrmann, T., Jones, P.D., Zeck, G., et al. (2022). A flexible protruding microelectrode array for neural interfacing in bioelectronic medicine. Microsyst. Nanoeng. *8*, 131. https://doi.org/10.1038/s41378-022-00466-z.

76. Ko, D.H., Bates, D., Karaosmanoglu, H., Taredun, K., Elton, C., Jones, L., Hosseini, A., and Partridge, A. (2023). 3D microelectrode arrays, pushing the bounds of sensitivity toward a generic platform for point-of-care diagnostics. Biosens. Bioelectron. *227*, 115154. https://doi.org/10.1016/j.bios.2023.115154.

77. Muzzi, L., Di Lisa, D., Falappa, M., Pepe, S., Maccione, A., Pastorino, L., Martinoia, S., and Frega, M. (2023). Human-Derived Cortical Neurospheroids Coupled to Passive, High Density and 3D MEAs: A Valid Platform for Functional Tests. Bioengineering *10*, 449. https://doi.org/10.3390/bioengineering10040449.

78. Leeson, H.C., Aguado, J., Gómez-Inclán, C., Chaggar, H.K., Fard, A.T., Hunter, Z., Lavin, M.F., Mackay-Sim, A., and Wolvetang, E.J. (2024). Ataxia Telangiectasia patient-derived neuronal and brain organoid models reveal mitochondrial dysfunction and oxidative stress. Neurobiol. Dis. *199*, 106562. https://doi.org/10.1016/j.nbd.2024.106562.

79. Lam, D., Fischer, N.O., and Enright, H.A. (2021). Probing function in 3D neuronal cultures: A survey of 3D multielectrode array advances. Curr. Opin. Pharmacol. *60*, 255–260. https://doi.org/10.1016/j.coph.2021.08.003.

80. Wilting, J., and Priesemann, V. (2018). Inferring collective dynamical states from widely unobserved systems. Nat. Commun. *9*, 2325. https://doi.org/10.1038/s41467-018-04725-4.

81. Shin, H., Jeong, S., Lee, J.-H., Sun, W., Choi, N., and Cho, I.-J. (2021). 3D high-density microelectrode array with optical stimulation and drug delivery for investigating neural circuit dynamics. Nat. Commun. *12*, 492. https://doi.org/10.1038/s41467-020-20763-3.

82. Soscia, D.A., Lam, D., Tooker, A.C., Enright, H.A., Triplett, M., Karande, P., Peters, S.K.G., Sales, A.P., Wheeler, E.K., and Fischer, N.O. (2020). A flexible 3-dimensional microelectrode array for *in vitro* brain models. Lab Chip *20*, 901–911. https://doi.org/10.1039/C9LC01148J.

83. Abu Shihada, J., Jung, M., Decke, S., Koschinski, L., Musall, S., Rincón Montes, V., and Offenhäusser, A. (2024). Highly Customizable 3D Microelectrode Arrays for In Vitro and In Vivo Neuronal Tissue Recordings. Adv. Sci. *11*, e2305944. https://doi.org/10.1002/advs.202305944.

84. Hayden, D.J., Finnie, P.S.B., Thomazeau, A., Li, A.Y., Cooke, S.F., and Bear, M.F. (2023). Electrophysiological signatures of visual recognition memory across all layers of mouse V1. J. Neurosci. *43*, 7307–7321. https://doi.org/10.1523/JNEUROSCI.0090-23.2023.

85. Le Floch, P., Li, Q., Lin, Z., Zhao, S., Liu, R., Tasnim, K., Jiang, H., and Liu, J. (2022). Stretchable Mesh Nanoelectronics for 3D Single-Cell Chronic Electrophysiology from Developing Brain Organoids. Adv. Mater. *34*, e2106829. https://doi.org/10.1002/adma.202106829.

86. Kalmykov, A., Reddy, J.W., Bedoyan, E., Wang, Y., Garg, R., Rastogi, S. K., Cohen-Karni, D., Chamanzar, M., and Cohen-Karni, T. (2021). Bioelectrical interfaces with cortical spheroids in three-dimensions. J. Neural Eng. *18*, 055005. https://doi.org/10.1088/1741-2552/abf290.

87. Cools, J., Jin, Q., Yoon, E., Alba Burbano, D., Luo, Z., Cuypers, D., Callewaert, G., Braeken, D., and Gracias, D.H. (2018). A Micropatterned Multielectrode Shell for 3D Spatiotemporal Recording from Live Cells. Adv. Sci. *5*, 1700731. https://doi.org/10.1002/advs.201700731.

88. Huang, Q., Tang, B., Romero, J.C., Yang, Y., Elsayed, S.K., Pahapale, G., Lee, T.-J., Morales Pantoja, I.E., Han, F., Berlinicke, C., et al. (2022). Shell microelectrode arrays (MEAs) for brain organoids. Sci. Adv. *8*, eabq5031. https://doi.org/10.1126/sciadv.abq5031.

89. Loeffler, A., Diaz-Alvarez, A., Zhu, R., Ganesh, N., Shine, J.M., Nakayama, T., and Kuncic, Z. (2023). Neuromorphic learning, working memory, and metaplasticity in nanowire networks. Sci. Adv. *9*, eadg3289. https://doi.org/10.1126/sciadv.adg3289.

90. Zhu, R., Hochstetter, J., Loeffler, A., Diaz-Alvarez, A., Nakayama, T., Lizier, J.T., and Kuncic, Z. (2021). Information dynamics in neuromorphic nanowire networks. Sci. Rep. *11*, 13047. https://doi.org/10.1038/s41598-021-92170-7.

91. Girardin, S., Clément, B., Ihle, S.J., Weaver, S., Petr, J.B., Mateus, J.C., Duru, J., Krubner, M., Forró, C., Ruff, T., et al. (2022). Topologically controlled circuits of human iPSC-derived neurons for electrophysiology recordings. Lab Chip *22*, 1386–1403. https://doi.org/10.1039/D1LC01110C.

92. Kriegman, S., Blackiston, D., Levin, M., and Bongard, J. (2020). A scalable pipeline for designing reconfigurable organisms. Proc. Natl. Acad. Sci. USA *117*, 1853–1859. https://doi.org/10.1073/pnas.1910837117.

93. Hattori, K., Kurakake, H., Imai, J., Hashimoto, T., Ishida, M., Sato, K., Takahashi, H., Oguma, S., Yamamoto, H., Hirano-Iwata, A., et al. (2021). Selective Stimulation of a Target Neuron in Micropatterned Neuronal Circuits Using a Pair of Needle Electrodes. Electrochemistry *89*, 348–354. https://doi.org/10.5796/electrochemistry.21-00032.

94. Orlandi, J.G., Soriano, J., Alvarez-Lacalle, E., Teller, S., and Casademunt, J. (2013). Noise focusing and the emergence of coherent activity in neuronal cultures. Nat. Phys. *9*, 582–590. https://doi.org/10.1038/nphys2686.

95. Ebitz, R.B., and Hayden, B.Y. (2021). The population doctrine in cognitive neuroscience. Neuron *109*, 3055–3068. https://doi.org/10.1016/j.neuron.2021.07.011.

96. Koroleva, A., Deiwick, A., El-Tamer, A., Koch, L., Shi, Y., Estévez-Priego, E., Ludl, A.-A., Soriano, J., Guseva, D., Ponimaskin, E., et al. (2021). *In Vitro* Development of Human iPSC-Derived Functional Neuronal Networks on Laser-Fabricated 3D Scaffolds. ACS Appl. Mater. Interfaces *13*, 7839–7853. https://doi.org/10.1021/acsami.0c16616.

97. Shein-Idelson, M., Ben-Jacob, E., and Hanein, Y. (2011). Engineered Neuronal Circuits: A New Platform for Studying the Role of Modular Topology. Front. Neuroeng. *4*, 10. https://doi.org/10.3389/fneng.2011.00010.

98. Sakaibara, M., Yamamoto, H., Murota, H., Monma, N., Sato, S., and Hirano-Iwata, A. (2024). Enhanced responses to inflammatory cytokine interleukin-6 in micropatterned networks of cultured cortical neurons. Biochem. Biophys. Res. Commun. *695*, 149379. https://doi.org/10.1016/j.bbrc.2023.149379.

99. Andolfi, A., Arnaldi, P., Lisa, D.D., Pepe, S., Frega, M., Fassio, A., Lagazzo, A., Martinoia, S., and Pastorino, L. (2023). A micropatterned thermoplasmonic substrate for neuromodulation of in vitro neuronal networks. Acta Biomater. *158*, 281–291. https://doi.org/10.1016/j.actbio.2022.12.036.

100. Jeong, S., Kang, H.W., Kim, S.H., Hong, G.-S., Nam, M.-H., Seong, J., Yoon, E.-S., Cho, I.-J., Chung, S., Bang, S., et al. (2023). Integration of reconfigurable microchannels into aligned three-dimensional neural networks for spatially controllable neuromodulation. Sci. Adv. *9*, eadf0925. https://doi.org/10.1126/sciadv.adf0925.

101. Lee, H., Yi, G.-S., and Nam, Y. (2023). Connectivity and network burst properties of in-vitro neuronal networks induced by a clustered structure with alginate hydrogel patterning. Biomed. Eng. Lett. *13*, 659–670. https://doi.org/10.1007/s13534-023-00289-5.

102. Xu, S., Liu, Y., Yang, Y., Zhang, K., Liang, W., Xu, Z., Wu, Y., Luo, J., Zhuang, C., and Cai, X. (2023). Recent Progress and Perspectives on Neural Chip Platforms Integrating PDMS-Based Microfluidic Devices and Microelectrode Arrays. Micromachines *14*, 709. https://doi.org/10.3390/mi14040709.

103. Mateus, J.C., Sousa, M.M., Burrone, J., and Aguiar, P. (2024). Beyond a Transmission Cable—New Technologies to Reveal the Richness in Axonal Electrophysiology. J. Neurosci. *44*, e1446232023. https://doi.org/10.1523/JNEUROSCI.1446-23.2023.

104. Rabadan, M.A., De La Cruz, E.D., Rao, S.B., Chen, Y., Gong, C., Crabtree, G., Xu, B., Markx, S., Gogos, J.A., Yuste, R., et al. (2022). An in vitro model of neuronal ensembles. Nat. Commun. *13*, 3340. https://doi.org/10.1038/s41467-022-31073-1.

105. Le Feber, J., Postma, W., De Weerd, E., Weusthof, M., and Rutten, W.L. C. (2015). Barbed channels enhance unidirectional connectivity between neuronal networks cultured on multi electrode arrays. Front. Neurosci. *9*, 412. https://doi.org/10.3389/fnins.2015.00412.

106. Winter-Hjelm, N., Brune Tomren, Å.B., Sikorski, P., Sandvig, A., and Sandvig, I. (2023). Structure-function dynamics of engineered, modular neuronal networks with controllable afferent-efferent connectivity. J. Neural Eng. *20*, 046024. https://doi.org/10.1088/1741-2552/ace37f.

107. Duru, J., Küchler, J., Ihle, S.J., Forró, C., Bernardi, A., Girardin, S., Hengsteler, J., Wheeler, S., Vörös, J., and Ruff, T. (2022). Engineered Biological Neural Networks on High Density CMOS Microelectrode Arrays. Front. Neurosci. *16*, 829884. https://doi.org/10.3389/fnins.2022.829884.

108. Sato, Y., Yamamoto, H., Kato, H., Tanii, T., Sato, S., and Hirano-Iwata, A. (2022). Microfluidic cell engineering on high-density microelectrode arrays for assessing structure-function relationships in living neuronal networks. Front. Neurosci. *16*, 943310. https://doi.org/10.3389/fnins.2022.943310.

109. Yamamoto, H., Moriya, S., Ide, K., Hayakawa, T., Akima, H., Sato, S., Kubota, S., Tanii, T., Niwano, M., Teller, S., et al. (2018). Impact of modular organization on dynamical richness in cortical networks. Sci. Adv. *4*, eaau4914. https://doi.org/10.1126/sciadv.aau4914.

110. Habibey, R., Striebel, J., Meinert, M., Latiftikhereshki, R., Schmieder, F., Nasiri, R., and Latifi, S. (2024). Engineered modular neuronal networks-on-chip represent structure-function relationship. Biosens. Bioelectron. *261*, 116518. https://doi.org/10.1016/j.bios.2024.116518.

111. Yamamoto, H., Spitzner, F.P., Takemuro, T., Buendía, V., Murota, H., Morante, C., Konno, T., Sato, S., Hirano-Iwata, A., Levina, A., et al. (2023). Modular architecture facilitates noise-driven control of synchrony in neuronal networks. Sci. Adv. *9*, eade1755. https://doi.org/10.1126/sciadv.ade1755.

112. Murota, H., Yamamoto, H., Monma, N., Sato, S., and Hirano-Iwata, A. (2025). Precision Microfluidic Control of Neuronal Ensembles in Cultured Cortical Networks. Adv. Mater. Technol. *10*, 2400894. https://doi.org/10.1002/admt.202400894.

113. Basso, G., and Barros, M.T. (2023). Biocomputing Model Using Tripartite Synapses Provides Reliable Neuronal Logic Gating With Spike Pattern Diversity. IEEE Trans. Nanobioscience *22*, 401–412. https://doi.org/10.1109/TNB.2022.3194834.

114. Borges, L.F., Barros, M.T., and Nogueira, M. (2024). Cell signaling error control for reliable molecular communications. Front. Commun. Netw. *5*, 1332379. https://doi.org/10.3389/frcmn.2024.1332379.

115. Amin, N.D., Kelley, K.W., Kaganovsky, K., Onesto, M., Hao, J., Miura, Y., McQueen, J.P., Reis, N., Narazaki, G., Li, T., et al. (2024). Generating human neural diversity with a multiplexed morphogen screen in organoids. Cell Stem Cell *31*, 1831–1846.e9. https://doi.org/10.1016/j.stem.2024.10.016.

116. McMillen, P., and Levin, M. (2024). Collective intelligence: A unifying concept for integrating biology across scales and substrates. Commun. Biol. *7*, 378. https://doi.org/10.1038/s42003-024-06037-4.

117. Watmuff, B., Habibollahi, F., Desouza, C., Khajehnejad, M., Loeffler, A., Baranes, K., Poulin, N., Kotter, M., and Kagan, B.J. (2025). Drug treatment alters performance in a neural microphysiological system of information processing. Commun. Biol. *8*, 916. https://doi.org/10.1038/s42003-025-08194-6.

118. Khajehnejad, M., Habibollahi, F., Loeffler, A., Paul, A., Razi, A., and Kagan, B.J. (2025). Dynamic Network Plasticity and Sample Efficiency in Neural Cultures: A Comparison with Deep Learning. Cyborg Bionic Syst. https://doi.org/10.34133/cbsystems.0336.

119. Boyd, J.L., and Lipshitz, N. (2024). Dimensions of Consciousness and the Moral Status of Brain Organoids. Neuroethics *17*, 5. https://doi.org/10.1007/s12152-023-09538-x.

120. Lavazza, A. (2021). Potential ethical problems with human cerebral organoids: Consciousness and moral status of future brains in a dish. Brain Res. *1750*, 147146. https://doi.org/10.1016/j.brainres.2020.147146.

121. Hyun, I., Scharf-Deering, J.C., and Lunshof, J.E. (2020). Ethical issues related to brain organoid research. Brain Res. *1732*, 146653. https://doi.org/10.1016/j.brainres.2020.146653.

122. Sawai, T., Hayashi, Y., Niikawa, T., Shepherd, J., Thomas, E., Lee, T.-L., Erler, A., Watanabe, M., and Sakaguchi, H. (2021). Mapping the Ethical Issues of Brain Organoid Research and Application. AJOB Neurosci. *13*, 81–94. https://doi.org/10.1080/21507740.2021.1896603.

123. Lavazza, A., and Reichlin, M. (2023). Human Brain Organoids: Why There Can Be Moral Concerns If They Grow Up in the Lab and Are Transplanted or Destroyed. Camb. Q. Healthc. Ethics, 1–15. https://doi.org/10.1017/S096318012300021X.

124. Sawai, T., Hayashi, Y., Niikawa, T., Shepherd, J., Thomas, E., Lee, T.-L., Erler, A., Watanabe, M., and Sakaguchi, H. (2022). Mapping the Ethical Issues of Brain Organoid Research and Application. AJOB Neurosci. *13*, 81–94. https://doi.org/10.1080/21507740.2021.1896603.

125. Birch, J., and Browning, H. (2021). Neural Organoids and the Precautionary Principle. Am. J. Bioeth. *21*, 56–58. https://doi.org/10.1080/15265161.2020.1845858.

126. Kagan, B.J., Loeffler, A., Boyd, J.L., and Savulescu, J. (2024). Embodied Neural Systems Can Enable Iterative Investigations of Morally Relevant States. J. Neurosci. *44*, e0431242024. https://doi.org/10.1523/JNEUROSCI.0431-24.2024.

127. Kagan, B.J., Duc, D., Stevens, I., and Gilbert, F. (2022). Neurons Embodied in a Virtual World: Evidence for Organoid Ethics? AJOB Neurosci. *13*, 114–117. https://doi.org/10.1080/21507740.2022.2048731.

128. Friston, K. (2023). The sentient organoid? Front. Sci. *1*, 1147911. https://doi.org/10.3389/fsci.2023.1147911.

129. Kataoka, M., Ishida, S., Kobayashi, C., Lee, T.-L., and Sawai, T. (2025). Evaluating neuroprivacy concerns in human brain organoid research. Trends Biotechnol. *43*, 491–493. https://doi.org/10.1016/j.tibtech.2024.09.001.

130. Lysaght, T. (2022). Anticipatory Governance and Foresight in Regulating for Uncertainty. Am. J. Bioeth. *22*, 51–53. https://doi.org/10.1080/15265161.2021.2001111.

131. Hasson, U., Nastase, S.A., and Goldstein, A. (2020). Direct Fit to Nature: An Evolutionary Perspective on Biological and Artificial Neural Networks. Neuron *105*, 416–434. https://doi.org/10.1016/j.neuron.2019.12.002.

132. Andersen, J., Revah, O., Miura, Y., Thom, N., Amin, N.D., Kelley, K.W., Singh, M., Chen, X., Thete, M.V., Walczak, E.M., et al. (2020). Generation of Functional Human 3D Cortico-Motor Assembloids. Cell *183*, 1913–1929.e26. https://doi.org/10.1016/j.cell.2020.11.017.

133. Miura, Y., Li, M.-Y., Revah, O., Yoon, S.-J., Narazaki, G., and Paşca, S.P. (2022). Engineering brain assembloids to interrogate human neural circuits. Nat. Protoc. *17*, 15–35. https://doi.org/10.1038/s41596-021-00632-z.

134. Roth, J.G., Brunel, L.G., Huang, M.S., Liu, Y., Cai, B., Sinha, S., Yang, F., Paşca, S.P., Shin, S., and Heilshorn, S.C. (2023). Spatially controlled construction of assembloids using bioprinting. Nat. Commun. *14*, 4346. https://doi.org/10.1038/s41467-023-40006-5.

135. Barabási, D.L., Ferreira Castro, A., and Engert, F. (2025). Three systems of circuit formation: assembly, updating and tuning. Nat. Rev. Neurosci. *26*, 232–243. https://doi.org/10.1038/s41583-025-00910-9.